

Example 4 : In a right triangle ABC, right-angled at B, if $\tan A = 1$, then verify that

$$2 \sin A \cos A = 1.$$

Solution : In $\triangle ABC$, $\tan A = \frac{BC}{AB} = 1$

$$\text{i.e., } BC = AB$$

Let $AB = BC = k$, where k is a positive number.

Now,

$$\begin{aligned} AC &= \sqrt{AB^2 + BC^2} \\ &= \sqrt{(k)^2 + (k)^2} = k\sqrt{2} \end{aligned}$$

$$\text{Therefore, } \sin A = \frac{BC}{AC} = \frac{k}{k\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\text{and } \cos A = \frac{AB}{AC} = \frac{k}{k\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\begin{aligned} \text{So, } 2 \sin A \cos A &= 2 \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = 1, \\ &\text{which is the required value.} \end{aligned}$$

Example 5 : In $\triangle OPQ$, right-angled at P, $OP = 7$ cm and $OQ = PQ = 1$ cm (see Fig. 8.12). Determine the values of $\sin Q$ and $\cos Q$.

Solution : In $\triangle OPQ$, we have

$$\begin{aligned} OQ^2 &= OP^2 + PQ^2 \\ (1 + PQ)^2 &= OP^2 + PQ^2 \quad (\text{Why?}) \\ 1 + PQ^2 + 2PQ &= OP^2 + PQ^2 \\ 1 + 2PQ &= 7^2 \quad (\text{Why?}) \\ PQ &= 24 \text{ cm and } OQ = 1 + PQ = 25 \text{ cm} \end{aligned}$$

$$\text{So, } \sin Q = \frac{7}{25} \text{ and } \cos Q = \frac{24}{25}$$

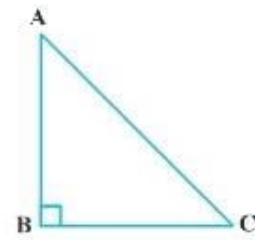


Fig. 8.11

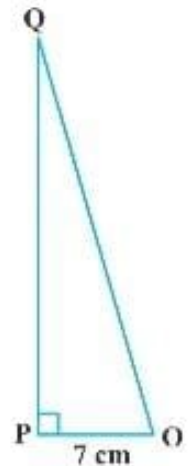


Fig. 8.12